

Navigators Technical Documentation

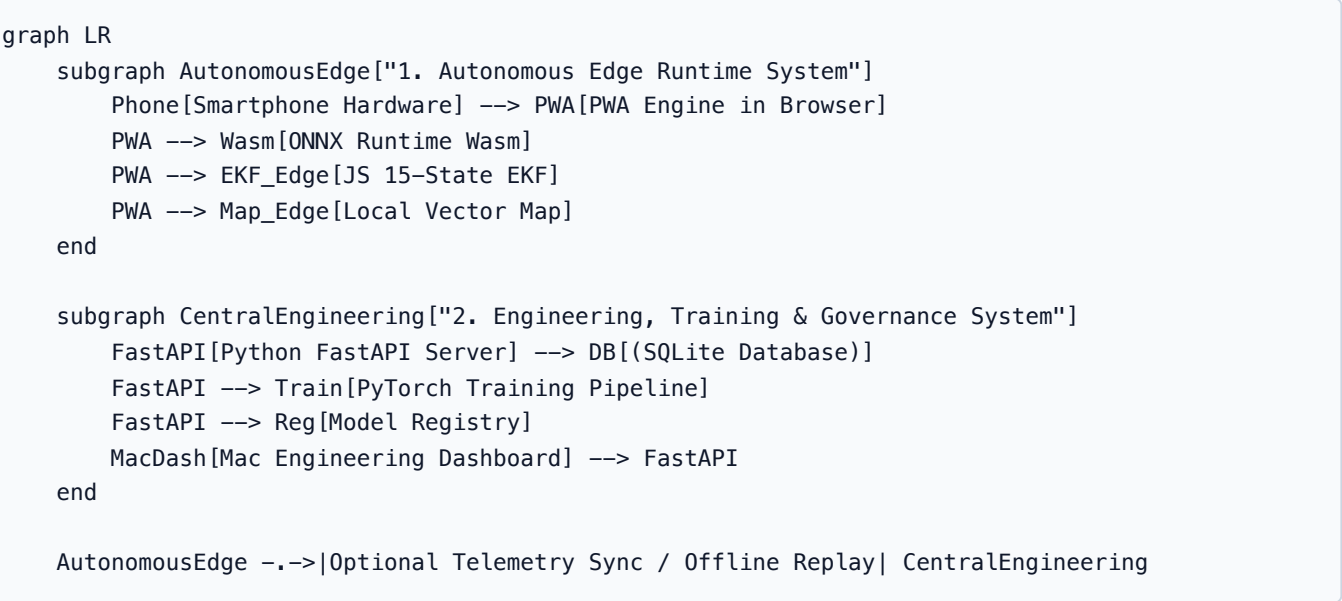
Navigators — Architecture Document

Document Identifier: ARCH-NAV-01
Status: Production Reference
Version: 1.0.0
Last Updated: 2026-09-23
Attribution: Developed by Navigators

1. Architectural Philosophy: Decoupled Systems

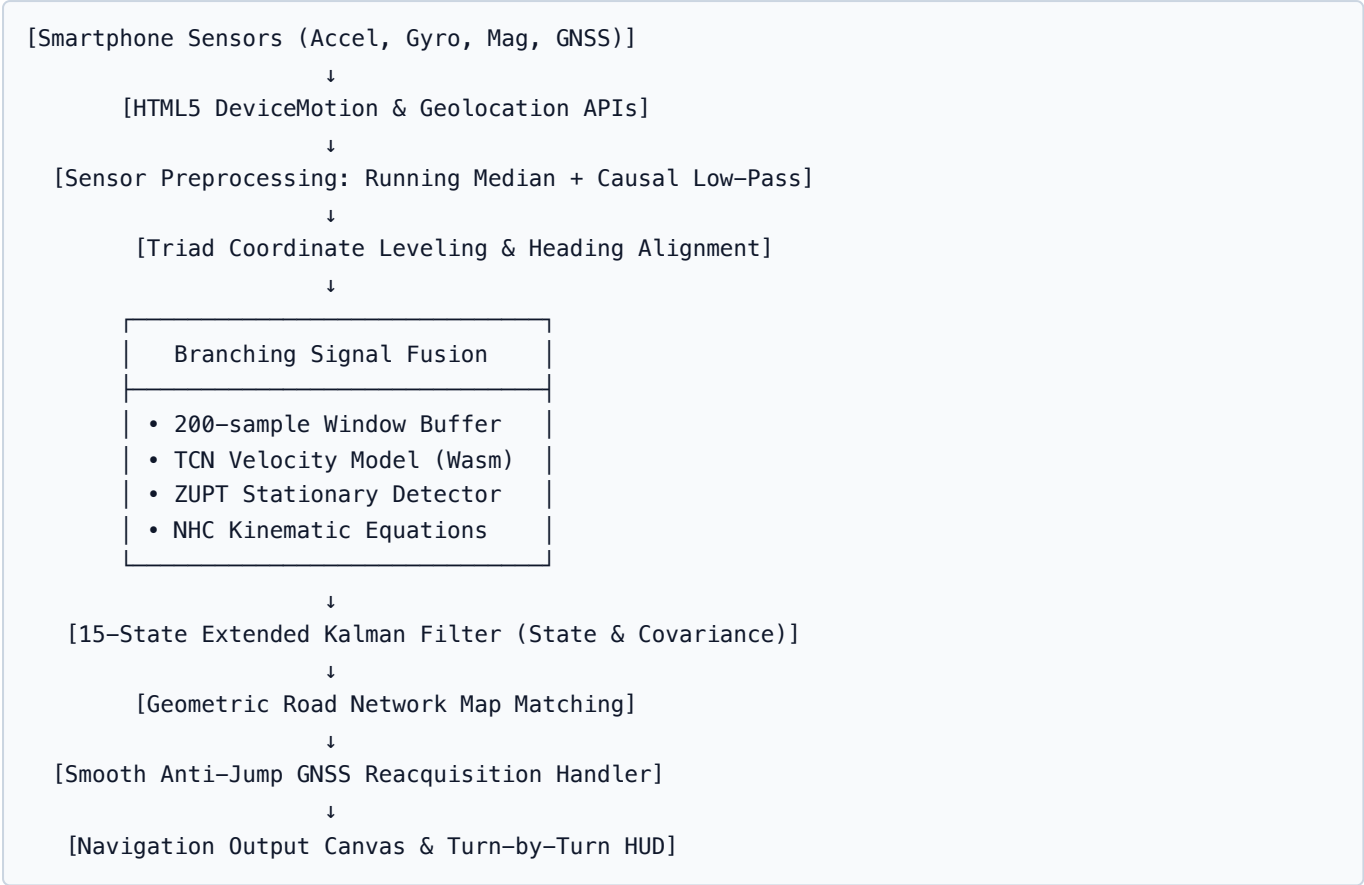
A critical architectural distinction in Navigators is the total separation between the **Client-Side Edge Navigation Runtime** and the **Server-Side Engineering & Governance System**:

[!IMPORTANT] **Zero Runtime Dependency:** The Mac Dashboard and Python FastAPI backend are **not runtime dependencies** for smartphone navigation. The client navigation application is entirely self-sufficient on-device once loaded into the browser or installed as a Progressive Web App (PWA).



2. Edge Navigation Runtime Architecture

The client navigation pipeline executes an unbroken dataflow running on the mobile device's local compute thread:

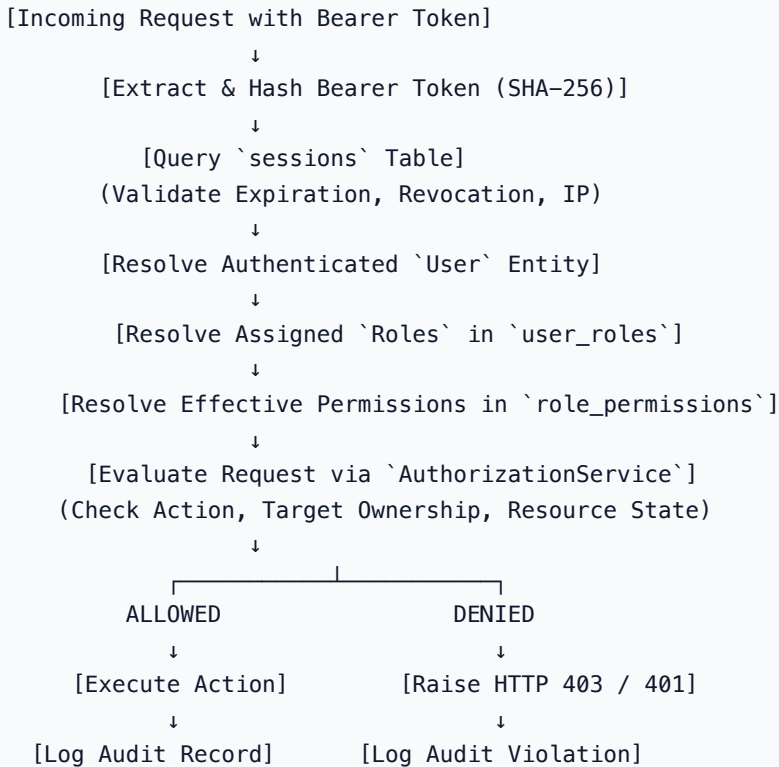


Detailed Pipeline Stages:

1. **Sensor Ingestion:** Captures tri-axial linear acceleration with gravity and angular rates.
2. **Causal Preprocessing:** Buffers events at 10 Hz. A running median filter removes spurious sensor spikes; a causal 1st-order Butterworth low-pass filter suppresses engine vibrations.
3. **Triad Frame Leveling:** Before motion begins, gravity vector averaging extracts the device pitch and roll angles relative to the Earth frame. Forward acceleration vector alignment resolves the vehicle yaw heading.
4. **AI Velocity Estimation:** A 200-sample sliding window of normalized features feeds the TCN model inside `ort-wasm-simd-threaded.wasm`, producing planar velocity predictions $[v_N, v_E]$.
5. **EKF Kinematic Fusion:** Predicts position propagation using high-rate IMU integration; corrects state estimates using AI velocity pseudo-measurements, Non-Holonomic Constraints ($v_y \approx 0, v_z \approx 0$), Zero Velocity Updates during stops, and GNSS measurements (when available).
6. **Map Matching:** Snaps unconstrained coordinates to OSM vector road centerlines stored in client memory.
7. **Anti-Jump Reacquisition:** When GNSS signals return after an extended outage, the filter caps step-wise corrections to ≤ 2 m per cycle over a 10-step horizon to eliminate visual position jumps.

3. Authentication, Authorization & Security Architecture

The server infrastructure enforces an unbroken chain of trust from incoming HTTP requests to immutable audit logging:

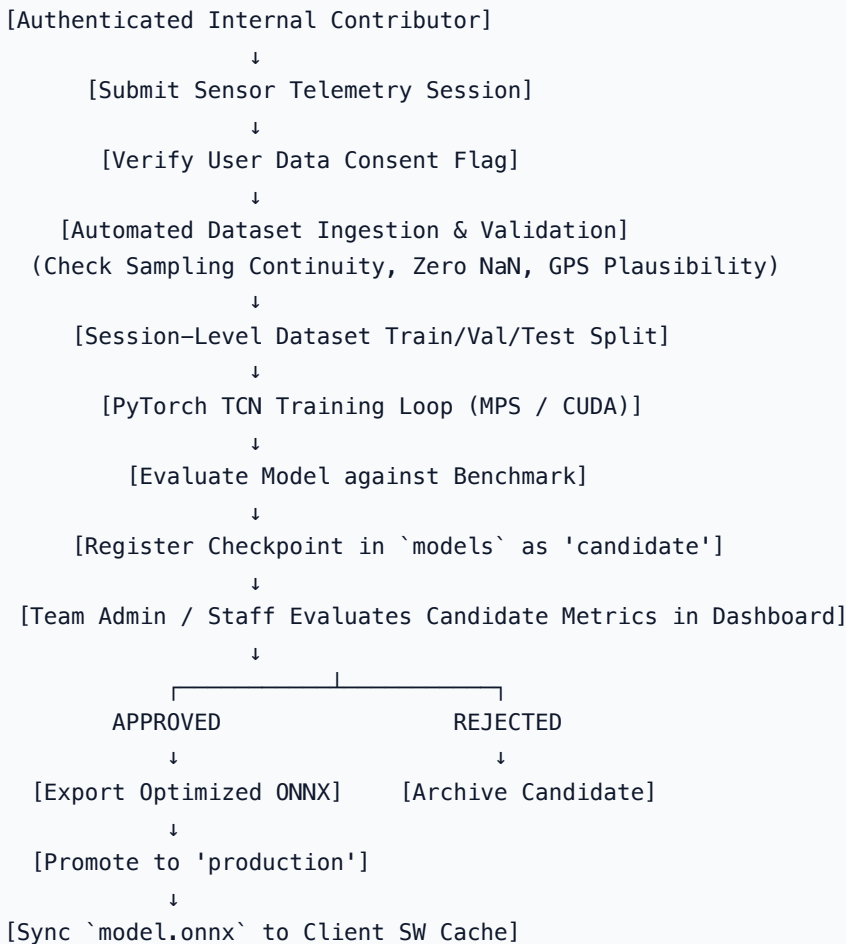


Key Security Guardrails:

- **Stateless Bearer Hashing:** Raw bearer tokens are never persisted in plaintext. Only the SHA-256 hash is recorded in the `sessions` table.
- **Hierarchical Roles:** 7 discrete roles are supported: `guest`, `user`, `local_contributor`, `internal_contributor`, `moderator`, `team_admin`, and `super_admin`.
- **State Machine Enforcement:** Role permissions alone cannot bypass entity states. For example, a Moderator cannot approve a contribution that is still in `draft` status.
- **Ownership Validation:** Authors possess exclusive rights to edit or withdraw their drafts; non-staff callers cannot view other users' unapproved contributions or private telemetry.

4. Engineering, Training & Model Lifecycle Architecture

The engineering pipeline provides an auditable, repeatable pathway from raw telemetry ingestion to production model deployment:



Governance Principles:

- **No Automatic Promotion:** Candidate models never replace production models automatically. Promotion requires explicit administrative action recorded in the audit trail.
- **Instant Rollback:** If a newly promoted model displays instability in field testing, administrators can trigger a rollback endpoint that reactivates the previous production checkpoint instantly.
- **Training Parity Verification:** Before deployment, the candidate ONNX artifact is tested against the original PyTorch model on identical input windows to verify numerical parity within $\text{atol} = 10^{-5}$, $\text{rtol} = 10^{-4}$.

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